

Onset of a quantum Fisher–Selke sequence in the transverse-field ANNNI model and its bilayer

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We study the transverse-field axial next-nearest-neighbor Ising (ANNNI) model on the square lattice and on a bilayer. At the classical multiphase point $\kappa = J_2/J_0 = 1/2$, every configuration of straight domain walls that separates domains of at least two sites is a ground state. The walls are rigid, so perturbation theory in the transverse field $g = \Gamma/J_0$ can be carried out for each configuration separately, and we take it to order g^8 . Quantum fluctuations select $\langle 3 \rangle$ ($\uparrow\uparrow\downarrow\downarrow$) at order g^2 , because spins next to a wall gain the most energy from virtual flips. The structures $\langle 23 \rangle$, $\langle 223 \rangle$ and $\langle 2223 \rangle$ then open between $\langle 3 \rangle$ and the antiphase $\langle 2 \rangle$ at orders g^4 , g^6 and g^8 , in the order of the Fisher–Selke sequence of the classical three-dimensional model. A second layer suppresses each successive phase more strongly, beyond the stiffening of the local fields. We test the two widest phases with stochastic series expansion quantum Monte Carlo, comparing the free energies of the competing ordered states by thermodynamic integration, since annealing freezes in the modulation that forms at the ordering transition. The simulations reproduce the $\langle 3 \rangle$ wedge to about 1%, and in a pre-registered test $\langle 23 \rangle$ is stable at six of seven points, with the seventh resolved by post-hoc runs.

I. INTRODUCTION

The axial next-nearest-neighbor Ising (ANNNI) model is the simplest lattice model in which competing ferromagnetic and antiferromagnetic interactions produce spatially modulated order [1, 2]. Along one axis, a ferromagnetic nearest-neighbor coupling J_1 competes with an antiferromagnetic next-nearest-neighbor coupling J_2 . At zero temperature the ferromagnet is stable for $\kappa = J_2/J_1 < 1/2$ and the antiphase $\langle 2 \rangle$, $\uparrow\uparrow\downarrow\downarrow$, for $\kappa > 1/2$, and at the multiphase point $\kappa = 1/2$ a large number of structures are degenerate. In the three-dimensional classical model, thermal fluctuations lift this degeneracy and generate an infinite sequence of commensurate phases that spring from the multiphase point [2–4]. In two dimensions the modulated region is instead dominated by a floating incommensurate phase with algebraic correlations [5, 6], whose extent was settled only recently [7–9].

A transverse field Γ adds quantum fluctuations. The transverse-field ANNNI chain is related to the classical two-dimensional model by the Suzuki–Trotter mapping [10] and has been studied in great detail [11–14]; the width of its floating phase was long disputed [12, 13]. In two and more dimensions the quantum ANNNI model has been treated by mean-field and random-phase approximations in three dimensions [15], by a path-integral estimate of the ferromagnetic boundary on the square and cubic lattices for $\kappa < 1/2$ [16], by large-spin perturbation theory for an anisotropic Heisenberg analog [17, 18], and, for quantum rotors, by field theory of

the quantum Lifshitz point [19]. To our knowledge there is no unbiased numerical study of the transverse-field ANNNI model with axial frustration on the square lattice or on its bilayer extension. Related frustrated transverse-field Ising models with diagonal couplings have been studied with quantum Monte Carlo [20–22] and, for bilayers, with cluster mean-field theory [23–25]; frustrated transverse-field Ising models are also realized on programmable quantum annealers [26].

Here we consider two square layers with axial frustration, coupled by an interlayer exchange, and ask two questions. What selects the ordered state near the multiphase point once quantum fluctuations are present, and how does the second layer change the answer? At the multiphase point of the square-lattice model, the in-plane ferromagnetic coupling perpendicular to the frustrated axis forces domain walls to be straight lines, so the degeneracy is subextensive and walls cannot move at low order in Γ . This makes the low-field problem accessible to perturbation theory. At second order in Γ we find that quantum fluctuations select the $\langle 3 \rangle$ structure $\uparrow\uparrow\downarrow\downarrow$ in a wedge around $\kappa = 1/2$ whose width grows as Γ^2 , and that the interlayer coupling narrows the wedge. At fourth order the structure $\langle 23 \rangle$ opens on the antiphase side of the wedge, at sixth order $\langle 223 \rangle$, and at eighth order $\langle 2223 \rangle$. This is the order of appearance in the low-temperature analysis of the classical three-dimensional model by Fisher and Selke [3, 27]. We establish these four steps and make no claim about how the sequence continues. The result differs both from the one-dimensional chain, where walls are mobile and the degeneracy gives way to a floating phase [11], and from the large-spin Heisenberg analog, where the phases $\langle 3 \rangle$, $\langle 4 \rangle$, $\langle 5 \rangle$, ... appear between the antiphase and the ferromagnet [17, 18].

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We test these predictions with stochastic series expansion (SSE) quantum Monte Carlo [28–30] at $T = 0.15 J_0$. Frustration makes the modulated region hard to equilibrate: cluster updates percolate at small field, and annealing from the paramagnet selects the modulation that forms at the ordering transition and cannot relax to a different commensurate structure at lower field. We therefore follow each competing ordered state as a separate branch and compare the branches by their free energies, obtained by thermodynamic integration in the field.

Up to $g/(2+r) \approx 0.5$ the free energies confirm the $\langle 3 \rangle$ wedge: its slope agrees with the prediction to about 1% for both r . In a test whose design and analysis were fixed before the simulations, $\langle 23 \rangle$ lies below both $\langle 2 \rangle$ and $\langle 3 \rangle$ at six of seven points. The seventh, with the smallest predicted margin, misses the pre-registered criterion; post-hoc runs with more statistics resolve $\langle 23 \rangle$ there as well. On a common lattice, at four points and with replicated runs in the bilayer, both edges of the window and both free-energy gains agree with the eighth-order series within their statistical errors. On the 20×30 lattices of the test, at the standard number of sweeps, the bilayer gain of $\langle 23 \rangle$ over $\langle 3 \rangle$ comes out about $10^{-4} J_0$ per spin larger; with four times the sweeps it agrees with the common lattice and with the series. The phases $\langle 223 \rangle$ and $\langle 2223 \rangle$ are too narrow to resolve numerically.

Section II defines the model and its symmetries. Section III derives the low-field phase diagram near the multiphase point. Section IV describes the SSE simulations, the branch protocol and their validation, and Sec. V presents the results. Section VI concludes.

II. MODEL

A. Hamiltonian

We consider two square layers $\ell = 1, 2$ of $L_x \times L_y$ sites (i, j) with periodic boundaries and $N = 2L_x L_y$ spins,

$$\begin{aligned} \hat{H} = & -J_x \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i+1,j,\ell}^z - J_y \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i,j+1,\ell}^z \\ & + J_2 \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i,j+2,\ell}^z - J_\perp \sum_{i,j} \hat{\sigma}_{i,j,1}^z \hat{\sigma}_{i,j,2}^z \\ & - \Gamma \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^x. \end{aligned} \quad (1)$$

The frustrated axis is y [Fig. 1(a)]. We set $J_x = J_y = J_0 > 0$ and use

$$\kappa = \frac{J_2}{J_0}, \quad g = \frac{\Gamma}{J_0}, \quad r_\perp = \frac{J_\perp}{J_0}. \quad (2)$$

The single-layer model is the case $r_\perp = 0$, in which the two layers decouple. The off-diagonal matrix elements $-\Gamma$ are nonpositive, so the model is stoquastic and has no sign problem in quantum Monte Carlo.

B. Symmetries

Two unitary transformations relate different couplings exactly. The layer flip $\hat{U}_2 = \prod_{i,j} \hat{\sigma}_{i,j,2}^x$ reverses $\hat{\sigma}^z$ in layer 2, changes the sign of the interlayer term and leaves all other terms invariant, so $\hat{H}(\kappa, g, r_\perp)$ and $\hat{H}(\kappa, g, -r_\perp)$ have the same spectrum and the same in-plane correlations; only the stacking of the two layers is reversed. The alternate-row spin flip $\hat{V} = \prod_{i,j \text{ odd}, \ell} \hat{\sigma}_{i,j,\ell}^x$ (for even L_y) reverses the sign of J_y and leaves J_x , J_2 and $J_\perp$ unchanged. We therefore take $J_y > 0$ and $r_\perp \geq 0$ without loss of generality, and write $r = |r_\perp|$.

C. Classical limit and ordered structures

At $g = 0$ the in-plane coupling J_x aligns neighboring spins along x , so in any low-energy configuration each column along y repeats in x , and the two layers are stacked in parallel for $r_\perp > 0$. The energy per spin is then fixed by the sequence of domains along y . For a structure in which all domains have at least two sites and the wall density along y is ρ (walls per site),

$$e_0 = -2 + \kappa - \frac{r}{2} + (2 - 4\kappa)\rho, \quad (3)$$

in units of J_0 . We denote a periodic structure by the lengths of the domains in one period, $\langle n_1 n_2 \dots \rangle$; thus $\langle 3 \rangle$ is $\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow$ and $\langle 23 \rangle$ is $\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow \dots$, which has period 5 and a net moment of $1/5$ per spin. The ferromagnet (FM, $\rho = 0$) has $e_0 = -2 + \kappa - r/2$, the antiphase $\langle 2 \rangle$ ($\rho = 1/2$) has $e_0 = -1 - \kappa - r/2$, and the structure $\langle 3 \rangle$, $\uparrow\uparrow\uparrow\downarrow\downarrow$ ($\rho = 1/3$), has $e_0 = -4/3 - \kappa/3 - r/2$. At $\kappa = 1/2$ the wall energy $2 - 4\kappa$ vanishes and every structure with domains of at least two sites is degenerate. Because walls must be straight lines along x , the number of such structures grows as $\exp(cL_y)$, not $\exp(cN)$: the degeneracy is subextensive.

At zero temperature, the Suzuki–Trotter mapping [10] relates the two-dimensional quantum model to an anisotropic three-dimensional classical ANNNI model. The thermal lifting of the multiphase degeneracy of the classical model is known from its low-temperature expansion [3, 27], but the mapped model lies outside the regime of that expansion (Sec. III C). At the temperature of our simulations the thermodynamics is two-dimensional. The ferromagnet then orders in the two-dimensional Ising universality class, and the structure $\langle n \rangle$, of period $2n$, carries a $\mathbb{Z}_{2n}$ clock-type order; for $2n \geq 6$ this order can melt through a floating phase [5, 6]. In mean-field theory the paramagnet becomes unstable at the wavevector $\cos q^* = 1/(4\kappa)$ for $\kappa > 1/4$ [2]. This is the wavevector of the instability at the ordering transition, not of the low-temperature ordered state.

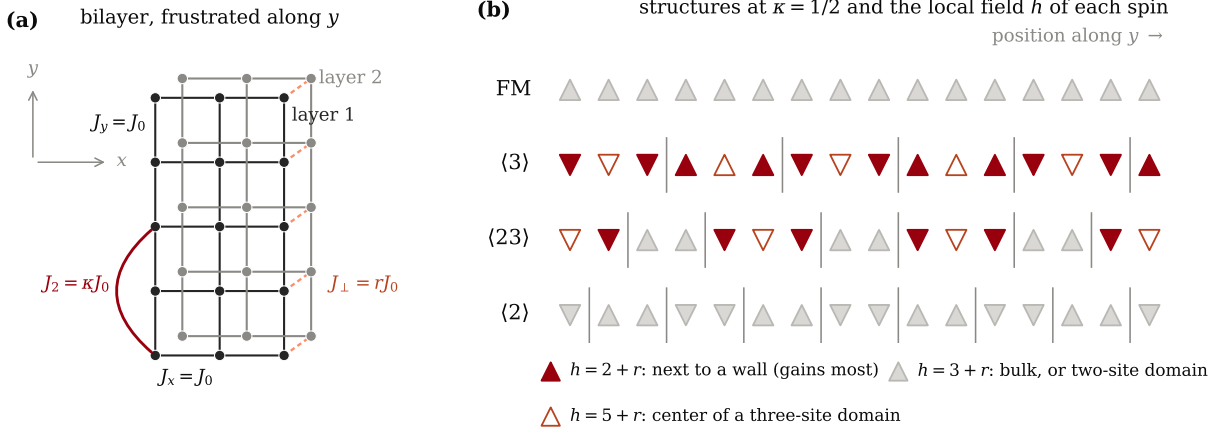


FIG. 1. (a) The bilayer of Eq. (1). Each layer is a square lattice with ferromagnetic couplings $J_x = J_y = J_0$ and an antiferromagnetic coupling $J_2 = \kappa J_0$ between next-nearest neighbors along the frustrated axis y (one such bond is drawn); the layers are coupled by $J_\perp = r J_0$. (b) Straight-wall structures along y at $\kappa = 1/2$: the ferromagnet, $\langle 3 \rangle$, $\langle 23 \rangle$ and the antiphase $\langle 2 \rangle$. Triangles give the spin direction and vertical lines the domain walls. Each spin is shaded by its local field h of Eq. (5) (Table I). Spins next to a wall ($h = 2 + r$) gain the most energy from virtual flips, spins in the bulk or in a two-site domain have $h = 3 + r$, and the center of a three-site domain has $h = 5 + r$.

D. Observables

We measure the energy per spin e , the transverse magnetization $m_x = N^{-1} \sum \langle \hat{\sigma}^x \rangle$, and equal-time zz correlations. Modulated order is characterized by the per-layer axial amplitudes

$$a_\ell(q) = \frac{1}{L_x L_y} \sum_{i,j} \sigma_{i,j,\ell}^z e^{iqj}, \quad q = \frac{2\pi n}{L_y}, \quad (4)$$

and the structure factor $S(q) = \frac{1}{2} \sum_\ell \langle |a_\ell(q)|^2 \rangle$. The dominant wavevector $q^* \in [0, \pi]$ maximizes $S(q)$. Since $a_\ell(-q) = a_\ell(q)^*$, the order parameter of a modulated state with $0 < q^* < \pi$ is $O(q^*) = 2S(q^*)$, and of the ferromagnet $O(0) = S(0)$. Both saturate at 1 for the ferromagnet and for $\langle 2 \rangle$, at $8/9$ for $\langle 3 \rangle$, and at $4(3+\sqrt{5})/25 \approx 0.838$ for $\langle 23 \rangle$. We normalize by these saturation values when comparing structures, and we identify the structure of a chain by q^* together with the correlations along y at distances 1 and 2. First-order transitions between ordered structures are located by crossings of branch free energies (Sec. IV B). The Binder cumulant [31] $U_4 = 1 - \langle m^4 \rangle / (3 \langle m^2 \rangle^2)$ and the second-moment correlation length ξ_y divided by the linear size are used only for the transverse-field Ising check of Sec. IV C.

III. LOW-FIELD PERTURBATION THEORY NEAR THE MULTIPHASE POINT

A. Local fields

For $g \ll 1$ and temperatures far below the excitation gaps, the free energy of an ordered structure is its en-

TABLE I. Local field h of Eq. (5) at $\kappa = 1/2$ for a spin in a domain of n sites, by its distance d from the nearer wall of the domain: $d = 1$ for a spin next to a wall, $d = 2$ for the next one, and so on.

Position	d	n	h
Next to a wall	1	≥ 3	$2 + r$
Second site from a wall	2	≥ 4	$4 + r$
Bulk	≥ 3	≥ 5	$3 + r$
Two-site domain	1	2	$3 + r$
Center of a three-site domain	2	3	$5 + r$

ergy (3) dressed by virtual single-spin flips. A flip of spin s costs $2hJ_0$, where h is the local field in units of J_0 . For a straight-wall structure with favorable stacking,

$$h = 2 + r + a_{+1} + a_{-1} - \kappa(a_{+2} + a_{-2}), \quad a_d = s_j s_{j+d}, \quad (5)$$

where j is the coordinate along y and the constant $2 + r$ collects the two in-plane neighbors along x and the partner spin in the other layer. At second order,

$$e_2 = -\frac{g^2}{2N} \sum_{\text{spins}} \frac{1}{h}. \quad (6)$$

At $\kappa = 1/2$, Eq. (5) takes five values, listed in Table I and illustrated in Fig. 1(b). Every spin with $d \geq 3$, and every spin of a two-site domain, has $h = 3 + r$. A spin next to a wall has the smaller field $2 + r$ and therefore gains more energy from quantum fluctuations. The field of a spin depends only on the length of its own domain, provided every domain has at least two sites.

B. The $\langle 3 \rangle$ wedge

Measured from the ferromagnet, the energy per spin of a sequence of domains of lengths n_j is

$$\Delta e = \frac{\sum_j [(2 - 4\kappa) - \frac{g^2}{2} F(n_j)]}{\sum_j n_j}, \quad (7)$$

where $F(n) = \sum_{\text{domain}} h^{-1} - n/(3 + r)$ is the excess second-order gain of one domain. From Table I,

$$\begin{aligned} F(2) &= 0, & F(3) &= \frac{6}{(2+r)(3+r)(5+r)}, \\ F(n \geq 4) &= F(3) - \frac{2}{(3+r)(4+r)(5+r)}. \end{aligned} \quad (8)$$

Because Eq. (7) is a ratio of sums, its minimum over all sequences is reached by a periodic structure $\langle n \rangle$ with a single domain length, the one that minimizes the per-domain cost divided by n , or by the ferromagnet if every domain costs a positive energy (Appendix A). Since $F(3) > F(n \geq 4)$ for every $r \geq 0$, no structure with domains longer than three is ever stable at this order. Comparing FM, $\langle 3 \rangle$ and $\langle 2 \rangle$ gives the phase boundaries

$$\kappa_{\text{FM}|3} = \frac{1}{2} - \frac{g^2}{8} F(3), \quad \kappa_{3|2} = \frac{1}{2} + \frac{g^2}{4} F(3). \quad (9)$$

Quantum fluctuations therefore open a wedge of $\langle 3 \rangle$ of width

$$\Delta\kappa = \frac{3g^2}{8} F(3) = \frac{9g^2}{4(2+r)(3+r)(5+r)} \quad (10)$$

that straddles $\kappa = 1/2$, with twice as much of it on the antiphase side. For decoupled layers, $F(3) = 1/5$ and the wedge is $1/2 - g^2/40 < \kappa < 1/2 + g^2/20$. For $r = 1$, $F(3) = 1/12$ and the wedge is $1/2 - g^2/96 < \kappa < 1/2 + g^2/48$, narrower by the factor $72/30 = 2.4$ at fixed g [Fig. 2(a),(b)]. The interlayer coupling adds the same constant r to every local field, which compresses the differences between $1/h$ values that drive the selection. Most of the factor reflects the larger local fields: in terms of $\lambda \equiv g/(2+r)$, the field in units of the local exchange scale, the wedge is $0.300\lambda^2$ wide for $r = 0$ and $0.281\lambda^2$ for $r = 1$, a ratio of 1.07. The selection of $\langle 3 \rangle$ from a degenerate manifold is an instance of order by disorder [32–34], here driven by quantum fluctuations, as in other frustrated transverse-field Ising models [35, 36]. These are the leading terms. At order g^4 (Sec. III C) the $\langle 3 \rangle$ wedge widens by $0.0161 g^4$ ($r = 0$) and $0.0024 g^4$ ($r = 1$), and above $\kappa = 1/2$ it ends at a new structure, $\langle 23 \rangle$, rather than at $\langle 2 \rangle$.

The two boundaries of Eq. (9) behave differently at higher order. At $\kappa_{\text{FM}|3}$ every domain other than a three-site one costs energy at order g^2 : a two-site domain costs $\frac{1}{2}g^2 F(3)$ and a domain of $n \geq 4$ sites $\frac{1}{2}g^2[F(3) - F(n)]$. Higher orders cannot overcome these costs at small g , so

the transition between the ferromagnet and $\langle 3 \rangle$ is a simple first-order level crossing. At $\kappa_{3|2}$, in contrast, every sequence of two- and three-site domains has the same energy at order g^2 , as in the thermal problem [3], and higher orders decide which of them are stable. Section III C shows that the fourth order stabilizes $\langle 23 \rangle$ there, and the sixth order $\langle 223 \rangle$.

C. Fourth, sixth and eighth order

At $\kappa_{3|2}$ every sequence of two- and three-site domains has the same energy at order g^2 , so higher orders decide which structures are stable there. We carry the expansion to order g^8 . Moving a wall takes a whole row of flips, so structures with different walls are coupled only at order g^{L_x} (g^{2L_x} for $r > 0$). The effective Hamiltonian in the manifold of straight-wall structures is therefore diagonal through order g^{L_x-1} (g^{2L_x-1} for $r > 0$), and its diagonal elements are the nondegenerate Rayleigh–Schrödinger energies of the individual structures. Taking $L_x \rightarrow \infty$ at fixed order, each straight-wall structure is a nondegenerate reference state. We write its energy per spin as

$$e = e_0 + \varepsilon_2 g^2 + \varepsilon_4 g^4 + \varepsilon_6 g^6 + \varepsilon_8 g^8 + O(g^{10}), \quad (11)$$

where $\varepsilon_2 g^2$ is the term e_2 of Eq. (6). Odd orders vanish because every flipped spin must be flipped back. Write the Ising part of Eq. (1) as $-\sum J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z$, so that $J_{ij} = -\kappa$ for the pairs at distance 2 along y . Let $D_i = 2h_i$ be the cost of flipping spin i , and let $B_{ij} = J_{ij} s_i s_j$, which is minus the energy of the bond between spins i and j . Then

$$N\varepsilon_4 = \sum_i \frac{1}{D_i^3} - \sum_{\langle ij \rangle} \frac{4B_{ij}(D_i + D_j)}{(D_i D_j)^2 (D_i + D_j - 4B_{ij})}. \quad (12)$$

The first term is the g^4 term of an isolated spin, whose energy is $-(h^2 + g^2)^{1/2}$. The second sum runs over coupled pairs only. For an uncoupled pair, the fourth-order processes that flip both spins cancel exactly against the normalization term of the expansion. This cancellation generalizes. Each spin that takes part in a process is flipped and restored, so the g^{2n} coefficient receives contributions only from connected clusters of at most n spins. We obtain ε_6 and ε_8 from this linked-cluster form, solving every connected cluster of up to three and four spins exactly in rational arithmetic. Equation (12) is exact; exact diagonalization of small clusters reproduces it to 3×10^{-6} relative, the accuracy of the fit to the diagonalization data. The sixth-order coefficient agrees with exact diagonalization to 10^{-4} relative, and with an independent evaluation that does not use the cluster decomposition to 10^{-15} . The eighth-order coefficient reproduces the exact series of the transverse-field Ising chain, $-25g^8/16384$ per spin, and equals a brute-force expansion over the full Hilbert space of 12-spin clusters exactly. An independent calculation that does not use the cluster decomposition

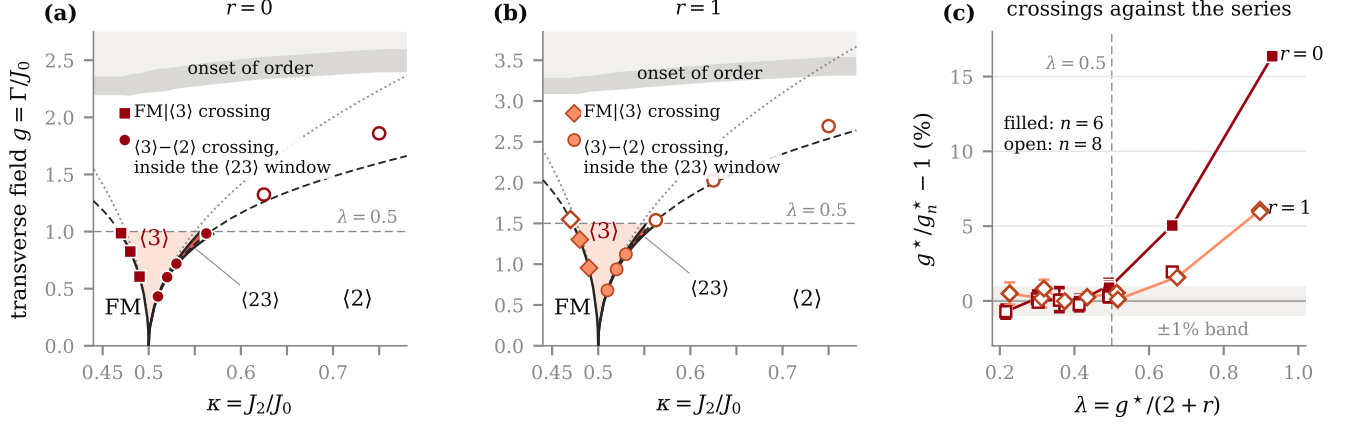


FIG. 2. Low-field phase diagram near the multiphase point for (a) $r = 0$ and (b) $r = 1$. Squares ($r = 0$) and diamonds ($r = 1$): fields g^* at which the $\langle 3 \rangle$ branch crosses the FM branch ($\kappa < 1/2$), a phase boundary. Circles: crossings of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches ($\kappa > 1/2$), which lie inside the $\langle 23 \rangle$ window and are not phase boundaries. Both are measured on 12×12 bilayers (Table IV); open symbols have $\lambda = g^*/(2+r) > 0.5$. Solid lines: phase boundaries at $O(g^6)$ at the actual κ , drawn up to $\lambda = 0.5$ (horizontal dashed line). The shaded region is $\langle 3 \rangle$ and the red band $\langle 23 \rangle$ (Sec. III C); the $\langle 223 \rangle$ window is too narrow to show. Dashed curves: $O(g^6)$ crossings of the $\langle 3 \rangle$ branch with the FM and $\langle 2 \rangle$ branches, which the symbols measure. Dotted curves: the $O(g^2)$ wedge, Eq. (9). Grey band: onset of order along the anneals from the paramagnet ($L = 12$), between the fields at which the mean $S(\pi/3)$ rises through 0.1 and 0.05. (c) Deviation of the measured crossings from the series at $O(g^6)$ (filled symbols) and $O(g^8)$ (open symbols) against λ ; the grey band is $\pm 1\%$. The $O(g^8)$ series has no crossing at $\kappa = 0.75$, $r = 0$. Error bars are one standard error; in (a) and (b) they are smaller than the symbols.

TABLE II. Coefficients of Eq. (11) per spin at $\kappa = 1/2$, in units of J_0 , for the structures that compete near $\kappa_{3/2}$; ρ is the wall density. $W_4 = k\varepsilon'_2 + \varepsilon_4$ with $k = F(3)/4$ [Eq. (14)] is the fourth-order energy on the $\langle 3 \rangle|\langle 2 \rangle$ line (not needed for FM). ε_2 and ε_4 are exact; W_4 , ε_6 and ε_8 are rounded (exact values are available with the data).

r		ρ	ε_2	ε_4	$10^3 W_4$	$10^5 \varepsilon_6$	$10^5 \varepsilon_8$
0	FM	0	-1/6	-5/1512	—	-24.15	-6.832
	$\langle 3 \rangle$	1/3	-1/5	-1/125	-7.333	-7.454	+60.27
	$\langle 23 \rangle$	2/5	-14/75	-54673/7560000	-4.610	-120.5	-30.38
	$\langle 2 \rangle$	1/2	-1/6	-13/4320	2.546	-22.06	-5.044
1	FM	0	-1/8	-1/1152	—	-2.062	-0.2020
	$\langle 3 \rangle$	1/3	-5/36	-437/362880	-1.011	4.409	-0.1433
	$\langle 23 \rangle$	2/5	-2/15	-4477/3628800	-0.5972	-2.175	-0.5922
	$\langle 2 \rangle$	1/2	-1/8	-11/13440	0.4836	-2.055	-0.1779

agrees with it to twelve digits. Table II lists the coefficients at $\kappa = 1/2$.

Expansion about the multiphase point. Write $\kappa = \frac{1}{2} + kg^2 + k_1g^4 + k_2g^6 + k_3g^8$. The classical energy (3) is linear in κ with slope $1 - 4\rho$, so Eq. (11) becomes

$$e = e_0|_{\kappa=1/2} + g^2[k(1-4\rho) + \varepsilon_2] + g^4[k_1(1-4\rho) + W_4] + g^6[k_2(1-4\rho) + W_6] + g^8[k_3(1-4\rho) + W_8] + O(g^{10}), \quad (13)$$

with the κ derivatives (primes) taken at $\kappa = 1/2$:

$$W_4 = k\varepsilon'_2 + \varepsilon_4, \quad W_6 = k_1\varepsilon'_2 + \frac{1}{2}k^2\varepsilon''_2 + k\varepsilon'_4 + \varepsilon_6, \\ W_8 = k_2\varepsilon'_2 + k k_1\varepsilon''_2 + \frac{1}{6}k^3\varepsilon'''_2 + k_1\varepsilon'_4 + \frac{1}{2}k^2\varepsilon''_4 + k\varepsilon'_6 + \varepsilon_8. \quad (14)$$

where W_6 and W_8 depend on the boundary considered through k_1 and k_2 (for instance k_1^- or k_1^+ of Eq. (15)). On the second-order boundary, $k = F(3)/4$, the g^2 bracket is the same for every sequence of two- and three-site domains (Appendix A). Each domain of $n \geq 4$ sites adds $\frac{1}{2}g^2[(n-2)F(3) - F(n)] > 0$ to the numerator of Eq. (7). At order g^4 only the sequences of 2s and 3s therefore compete, with energy $k_1(1-4\rho) + W_4$ per spin. Since the first term is linear in ρ , the stable structures are the vertices of the lower convex hull of W_4 plotted against ρ . The same construction with W_6 and W_8 decides the boundaries that remain degenerate at the preceding order. The κ derivatives are obtained exactly by carrying $\kappa = \frac{1}{2} + \delta$ through the cluster sums as a truncated power series in δ .

Fourth order: the $\langle 23 \rangle$ phase. At $\kappa = 1/2$ the field (5) of a spin depends only on its position in its own domain (Table I). The pairs in Eq. (12) lie either within one domain or on the two sides of a single wall. W_4 is therefore a sum of terms for single domains and for pairs of adjacent domains [Eq. (A2)]. A periodic sequence can be read as a closed path through the domain lengths, and its energy per spin is the cost of the path divided by its length. Such a ratio is smallest on an elementary cycle, so only $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$ ($\uparrow\uparrow\downarrow\downarrow$) can be stable at this

TABLE III. Coefficients of the phase boundaries near $\kappa = 1/2$. The edges of the $\langle 23 \rangle$ window are $\kappa = \frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^\pm g^4 + k_2^\pm g^6$ (from sixth order the edge on the $\langle 2 \rangle$ side is the $\langle 23 \rangle|\langle 223 \rangle$ boundary); k_1^{32} and k_1^F belong to the crossing of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches and to the FM| $\langle 3 \rangle$ boundary. $\Delta\kappa_s$ is the width of the window of structure s at the order at which it opens, and $\delta_{233}^{(6)}g^6 + \delta_{233}^{(8)}g^8$ the height of $\langle 233 \rangle$ per spin above the segment from $\langle 3 \rangle$ to $\langle 23 \rangle$. Decimal entries are rounded.

	$r = 0$	$r = 1$
k_1^-	20591/2016000	167/107520
k_1^+	6011/336000	1961/725760
k_1^{32}	1067/72000	31/13824
k_1^F	-59/10080	-1573/1935360
$10^3 k_2^-$	-3.06	-0.143
$10^3 k_2^+$	+5.26	+0.242
$\Delta\kappa_{23}/g^4$	7.68×10^{-3}	1.15×10^{-3}
$\Delta\kappa_{223}/g^6$	4.1×10^{-4}	2.3×10^{-5}
$\Delta\kappa_{2223}/g^8$	1.8×10^{-5}	3.8×10^{-7}
$\delta_{233}^{(6)}$	$+2.0 \times 10^{-6}$	$+1.2 \times 10^{-7}$
$\delta_{233}^{(8)}$	-9.7×10^{-7}	-1.8×10^{-8}

order, among sequences of any length (Appendix A). Table II places $\langle 23 \rangle$ strictly below the chord from $\langle 3 \rangle$ to $\langle 2 \rangle$, so $\langle 23 \rangle$ is stable in a window

$$\frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^- g^4 < \kappa < \frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^+ g^4, \quad (15)$$

with the coefficients of Table III. Its width, $\Delta\kappa_{23} = (k_1^+ - k_1^-)g^4$, is 6.7 times smaller in the bilayer than in the single layer at fixed g , against 2.4 for the $\langle 3 \rangle$ wedge. Most of this difference is power counting, $(3/2)^4$ against $(3/2)^2$: at fixed λ the two ratios are 1.32 and 1.07. At fixed λ the window still narrows steadily with r , from $0.123\lambda^4$ at $r = 0$ to $0.107\lambda^4$, $0.093\lambda^4$ and $0.071\lambda^4$ at $r = 1/2, 1$ and 2 . All these factors are ratios of asymptotic coefficients. The metastable branches $\langle 3 \rangle$ and $\langle 2 \rangle$ cross inside the window, at $\kappa = \frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^{32}g^4$, the line along which their template free energies can be compared with simulations. On the ferromagnetic side the boundary moves to $\kappa_{\text{FM}|\langle 3 \rangle} = \frac{1}{2} - \frac{1}{8}F(3)g^2 + k_1^F g^4$, and no new structure appears there: every other structure contains two-site domains or domains of four or more sites, each of which costs energy at order g^2 that higher orders cannot overturn at small g .

Degenerate edges. The argument also shows exactly which structures tie on the window edges. In the (ρ, W_4) plane, a sequence with no two adjacent three-site domains ($\langle 223 \rangle$, $\langle 2223 \rangle$, $\langle 22323 \rangle$, ...) lies exactly on the segment joining $\langle 2 \rangle$ and $\langle 23 \rangle$. A sequence with no two adjacent two-site domains ($\langle 233 \rangle$, $\langle 2333 \rangle$, ...) lies exactly on the segment joining $\langle 23 \rangle$ and $\langle 3 \rangle$ (Appendix A). Each edge of the window is therefore itself a multiphase line at order g^4 , and the next order decides it.

Sixth order. A connected cluster of three spins spans at most one intermediate domain. W_6 is therefore a sum of terms for three consecutive domains, which we have

confirmed in exact arithmetic. Applied to pairs of consecutive domains, the elementary-cycle argument allows at most one new structure in each gap at this order: $\langle 223 \rangle$ between $\langle 23 \rangle$ and $\langle 2 \rangle$, or $\langle 233 \rangle$ between $\langle 3 \rangle$ and $\langle 23 \rangle$. $\langle 223 \rangle$ opens (Table III). $\langle 233 \rangle$ does not, but it lies above the segment from $\langle 3 \rangle$ to $\langle 23 \rangle$ by only about $2 \times 10^{-6}g^6$ per spin at $r = 0$ (Table III), so the edge is only weakly first order; beyond the range of the series the simulations find $\langle 233 \rangle$ and $\langle 2333 \rangle$ below both (Sec. V C). Every other mixture of the two ($\langle 2333 \rangle$, $\langle 23233 \rangle$ and longer ones) contains at least one triple $(2, 3, 3)$ or $(3, 3, 2)$ of consecutive domains and also lies above the segment, so the $\langle 3 \rangle|\langle 23 \rangle$ boundary is of first order at small g .

Eighth order. At order g^8 the energy is a sum of terms for four consecutive domains. On the $\langle 223 \rangle|\langle 2 \rangle$ line $\langle 2223 \rangle$ lies below the chord from $\langle 223 \rangle$ to $\langle 2 \rangle$ and opens a narrow window (Table III). Longer mixtures such as $\langle 22223 \rangle$ lie on the new segments, so $\langle 2223 \rangle$ is the only structure that enters this line at this order. On the $\langle 23 \rangle|\langle 23 \rangle$ line the elementary cycles are $\langle 23 \rangle$ and $\langle 223 \rangle$ alone (Appendix A), so no structure enters there at this order; mixtures such as $\langle 23223 \rangle$ remain exactly degenerate, and a higher order decides that edge. On the $\langle 3 \rangle|\langle 23 \rangle$ line the eighth order lowers the margin of $\langle 233 \rangle$, but the truncated series would change its sign only at $g \approx 1.42$ ($r = 0$) and 2.60 ($r = 1$), beyond the range in which it converges (see below). Through order g^8 the transverse field thus selects $\langle 3 \rangle$, $\langle 23 \rangle$, $\langle 223 \rangle$ and $\langle 2223 \rangle$, in this order, between the ferromagnet and the antiphase. A structure $\langle 2^k 3 \rangle$ can first appear at order $g^{2(k+1)}$, the lowest order at which a connected cluster spans the $k+1$ domains of its period, and it does so for $k \leq 3$. Whether it does so for every k depends on the sign of each new coefficient; in the classical expansion, corrections at large k make the sequence finite at any nonzero temperature and admit mixed phases [2, 37]. Higher orders would decide this here (Sec. VI).

The same order of appearance occurs in the low-temperature expansion of the classical three-dimensional model [2, 3, 27]. There $\langle 3 \rangle$ appears at first order in the expansion parameter with a first-order boundary to the ferromagnet, $\langle 23 \rangle$ at second order, and each $\langle 2^k 3 \rangle$ at order $k+1$, the number of connected spins that must flip to span the sequence [37]. In the language of domain-wall interactions [38–40], g^2 takes over the role of the low-temperature expansion parameter. The selecting excitations are virtual rather than thermal spin flips next to the walls, and the coefficients are different. One difference is structural. In the classical calculation to second order, the $\langle 3 \rangle|\langle 23 \rangle$ transition is of first order, while on the $\langle 23 \rangle|\langle 2 \rangle$ boundary all sequences of two-bands and three-bands surrounded by two-bands remain degenerate [27][2, Sec. 4.2]; here both edges of the $\langle 23 \rangle$ window are degenerate at $O(g^4)$, and the $\langle 3 \rangle|\langle 23 \rangle$ edge becomes first order only at $O(g^6)$. At $T = 0$ the quantum model maps onto a strongly anisotropic three-dimensional classical model [10], outside the regime of the classical low-temperature expansion. The classical boundary between

the ferromagnet and $\langle 3 \rangle$ has been derived rigorously at low temperature [41], and quantum Pirogov–Sinai theory controls quantum perturbations of classical ground states, including infinitely degenerate ones [42, 43]. Our statements are asymptotic in the same sense: each holds at its order for sufficiently small g , with $L_x \rightarrow \infty$ taken first (Sec. III D).

The sixth order also shifts the edges of the $\langle 23 \rangle$ window (Table III) and widens it by a relative amount $1.08 g^2$ at $r = 0$ and $0.34 g^2$ at $r = 1$, as large as the leading term near $g = 1$ for $r = 0$. The fourth-order width is therefore quantitative only at small g . At $r = 0$ the eighth order is again large: ε_8 of $\langle 3 \rangle$ is eight times ε_6 in magnitude, with opposite sign (Table II).

Crossings at finite $\kappa - 1/2$. For fixed $\kappa > 1/2$ the leading-order result, Eq. (9), places the crossing of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches at $g_{3|2} \simeq [4(\kappa - 1/2)/F(3)]^{1/2}$ for κ close to $1/2$. Away from $\kappa = 1/2$ the crossing is better obtained from Eq. (11) evaluated at the actual κ . Table IV lists these crossings at each order next to the simulations (Sec. V A).

Convergence. The spins next to a wall of a domain of three or more sites feel no field from the couplings along y , at any κ [Eq. (5)]: their field $2 + r$ comes from the neighbors along x and from the partner in the other layer. With its environment frozen, the row next to a wall is therefore a transverse-field Ising chain for $r = 0$ and a two-leg ladder with equal leg and rung couplings for $r = 1$. Exact diagonalization of chains of up to 16 sites and ladders of up to 10 rungs places their critical fields at $g = 1.00$ and 1.83 , that is, at $\lambda = 0.50$ and 0.61 . The estimate ignores the weak pinning of the rows by the wall interactions, so it identifies a scale rather than a radius of convergence, but the series for structures with such rows is expected to deteriorate near these fields, where the rows next to the walls, and with them the walls, soften. This is the quantum counterpart of the wall delocalization that destabilizes $\langle 3 \rangle$ in the classical model [2]. A breakdown of this kind need not show up in the low-order coefficients: for $r = 1$ at $\kappa = 0.625$ and 0.75 the $O(g^6)$ and $O(g^8)$ crossings nearly coincide, yet the simulated crossings, which lie beyond the ladder scale, are well above both (Table IV). The edge rows set the range $\lambda \lesssim 0.5$ ($r = 0$) and $\lambda \lesssim 0.6$ ($r = 1$) in which Sec. V compares the series with the simulations.

D. Why the square lattice differs from the chain

The perturbative argument relies on the walls being rigid. On the square lattice, moving a wall by one site requires flipping a whole row of L_x spins (both rows, $2L_x$ spins, in the bilayer with $r > 0$), which enters only at order g^{L_x} (g^{2L_x}), so the walls are frozen at any finite order. In the chain ($J_x = 0$, $r = 0$) the field next to a wall of a domain of three or more sites vanishes by Eq. (5); at $\kappa = 1/2$, where walls cost nothing, a single flip then moves a wall at no cost, and the walls

delocalize, as described by effective domain-wall Hamiltonians for the transverse chain [11]. The degeneracy is then lifted already at first order in g , and a gapless floating region opens between two lines linear in g , $-g/2 < \kappa - 1/2 < g$ [12], in place of the commensurate structures found here. The large-spin analysis of an anisotropic Heisenberg model on the cubic lattice [17, 18] also finds that quantum fluctuations lift the degeneracy. There, however, the selecting excitations are spin deviations that propagate through the transverse exchange. The resulting phases, $\langle 3 \rangle$, $\langle 4 \rangle$, $\langle 5 \rangle$, $\dots$, lie between the antiphase and the ferromagnet with the period increasing toward the ferromagnet, and the $\langle 2 \rangle|\langle 3 \rangle$ boundary remains of first order, with no $\langle 23 \rangle$ phase. At the simulation temperature $T = 0.15 J_0$, thermal steps on a wall cost at least $2J_0$ each, and on a periodic wall they come in pairs; even the conservative factor $e^{-2J_0/T} \approx 10^{-6}$ keeps the walls straight on the scale of our lattices. Virtual flips of the rows next to the walls are the relevant quantum fluctuations, and they limit the convergence of the series (Sec. III C). As these rows approach their critical field their kink energy falls, and thermal kinks, which do not move the walls, add to the free energy; Sec. S3 of the Supplemental Material [44] estimates this contribution. The data agree. At $g_0 = 0.05$ the nearest-neighbor correlation along x lies within 6×10^{-4} of unity on every template branch, the reduction expected from virtual single flips alone ($4g_0^2 \langle D^{-2} \rangle$ at second order). Steps on the walls would lower it further, and they are not resolved.

The choice $J_x = J_y$ is not essential. For $J_x \neq J_y$, with energies in units of J_y , the constant $2 + r$ in Eq. (5) becomes $2J_x/J_y + r$, so every local field shifts by $2(J_x/J_y - 1)$, and the second-order results, Eqs. (9) and (10), hold with r replaced by $r + 2(J_x/J_y - 1)$. At higher orders J_x also enters the couplings between flipped spins, so the coefficients change, but the rigidity of the walls for any $J_x > 0$, the dependence of each local field on its own domain only, and the elementary-cycle argument do not; the same candidates arise at each order, and whether they open depends on the coefficients. For $r = 0$ the rows next to the walls become critical at $\Gamma \approx J_x$, so the range of the series shrinks as $J_x \rightarrow 0$, toward the chain limit described above, where the walls move freely.

IV. METHODS

A. Stochastic series expansion

We write Eq. (1) as $\hat{H} = -\sum_b \hat{H}_b - \sum_s (\hat{H}_{s,c} + \hat{H}_{s,f}) + C$, with bond operators $\hat{H}_b = |J_b| + J_b \hat{\sigma}_i^z \hat{\sigma}_j^z$, where J_b is the coupling of bond b in the convention $-J_b \hat{\sigma}_i^z \hat{\sigma}_j^z$, a constant site operator $\hat{H}_{s,c} = \Gamma$, a spin-flip operator $\hat{H}_{s,f} = \Gamma \hat{\sigma}_s^x$, and $C = \sum_b |J_b| + NT$. All matrix elements are nonnegative for either sign of J_b , so frustration affects only the efficiency. Diagonal updates insert and remove $\hat{H}_b$ and $\hat{H}_{s,c}$ in the operator string, and cluster updates build Swendsen–Wang-type clusters that pass through

bond operators and end on site operators [29]. The energy is $e = (-\langle n \rangle / \beta + C) / N$ with n the number of operators, the transverse magnetization is $m_x = \langle n_f \rangle / (\beta \Gamma N)$ with n_f the number of flip operators, and the nearest-neighbor, next-nearest-neighbor and interlayer correlations follow from the numbers of bond operators of each type. Equal-time quantities are measured on the propagated state at imaginary time zero. There is no Trotter error. A Monte Carlo sweep consists of one diagonal update of the whole operator string and one cluster update, in which every cluster is flipped with probability 1/2. The string length M grows during thermalization, to $n + n/2 + 64$ whenever $n > 3M/4$, and is fixed during measurements. Unless stated otherwise, every chain performs 2000 thermalization and 3000 measurement sweeps at each field, with measurements averaged in bins of 100 sweeps. At the test field of Sec. VB ($\kappa = 0.53$, $r = 1$, $g_c = 1.12$) the integrated autocorrelation times, estimated from bins of 10 sweeps, are about 6 sweeps for the energy and 12–17 sweeps for m_x , the same for $\langle 3 \rangle$ and $\langle 23 \rangle$ on 12×12 , 20×30 and 20×60 lattices. A field thus provides about 200 independent values of m_x per chain, and the 2000 thermalization sweeps exceed these times a hundredfold. Near the crossover of the 12-spin cluster of Sec. IV C the time for m_x grows to about 50 sweeps.

B. Equilibration in the modulated region

At small g the operator string holds few site operators, the clusters of satisfied bonds percolate, and the cluster update reduces to global flips, so local moves cannot convert one ordered structure into another. On 12-site clusters, annealing downward from the paramagnet removes these traps. On larger lattices it does not. Figure 3(a) shows an anneal on a 12×12 bilayer at $\kappa = 0.75$: $\langle 3 \rangle$ order forms at the ordering transition and persists below the field g^* at which $\langle 2 \rangle$ becomes lower, so at $r = 0$ the chains end $0.17J_0$ per spin above the stable antiphase. Of 138 chains annealed to $g_0 = 0.05$ in the runs of Sec. VA, at κ between 0.47 and 0.75 and on 12×12 and 24×24 lattices, 137 end in $\langle 3 \rangle$, even for $\kappa < 1/2$ where the ferromagnet is stable; the remaining one ends in another sequence with the same wall density. Annealing ten times more slowly leaves this unchanged on 12×12 lattices, where only $\langle 2 \rangle$ and $\langle 3 \rangle$ fit: at $\kappa = 0.75$ all twelve chains, six for each r , again end in $\langle 3 \rangle$.

This freezing is set by the wavevector at which order forms. On $L_y = 12$ the allowed wavevector nearest the mean-field instability, $\cos q^* = 1/(4\kappa)$, is $\pi/3$ for every κ studied, and only $\langle 2 \rangle$ and $\langle 3 \rangle$ fit among sequences of two- and three-site domains. On 12×60 lattices, which resolve the wavevectors in between, order forms instead at the allowed wavevector nearest the instability and stays there [Fig. 3(b)]; at $\kappa = 0.625$ and 0.75 every chain ends in a sequence of two- and three-site domains with that wall density, none in $\langle 3 \rangle$. In separate low-field checks on 24×24

lattices, anneals from $g = 3$ ($r = 0$) or 5 ($r = 1$) also leave frozen stripes at $\kappa = 0.25$, as in zero-temperature quenches of Ising ferromagnets [45], and frozen domain-wall configurations for $\kappa \geq 0.75$. Exchange Monte Carlo in the field [46] could reduce the freezing; we do not pursue it, because the template branches below give the free energies directly.

We therefore treat each candidate ordered structure as a separate branch. Chains are started in the FM, $\langle 2 \rangle$ and $\langle 3 \rangle$ structures at g_0 and stepped upward in g . Along each branch the free energy per spin follows from $\partial F / \partial \Gamma = -\langle \sum_s \hat{\sigma}_s^x \rangle$,

$$f_b(g) = f_b(g_0) - \int_{g_0}^g m_x^{(b)}(g') dg'. \quad (16)$$

At g_0 the entropy of an ordered state is negligible (the lowest excitation costs $2(2+r)J_0$), so we take $f_b(g_0)$ from the perturbation series (11) of the template; the measured energies at g_0 agree with it [Fig. 6(b)], and using the series instead of the measured starting point reduces the uncertainty of the crossings by a median factor of 3.6; crossings from the measured starting points agree within 2σ . The branch free energy is that of one fixed arrangement of walls. The entropy of the many arrangements with similar energies, of order $1/L_x$ per spin, is not sampled because the walls are rigid. It could reach $10^{-3}J_0$ per spin on our lattices, so the Gibbs state of a finite lattice would mix arrangements, but it vanishes in the thermodynamic limit, where the branches are the appropriate objects to compare. Before the first cluster update the operator string is filled by diagonal updates alone, so that the initial structure survives (Appendix B).

The integral is valid while a branch remains in one structure, so we cut a chain at the first change of its dominant wavevector; every crossing lies at least 0.24 below the first cut, and truncating the branches there moves the crossings of Table IV by at most 3×10^{-4} . We integrate m_x with the trapezoidal rule and a Richardson correction, locate crossings by a cubic-spline root, and estimate errors by the jackknife over independent chains, with Student- t intervals; these choices were fixed after a first analysis of the data of Sec. VA and before the test of Sec. VB. The stable state at each g is the branch of lowest f_b among the structures compared, and a crossing of two branches locates a first-order transition; how far a branch persists beyond it depends on the lattice and the update and is not a thermodynamic hysteresis. As a check of the integration, the measured energy e_b must equal f_b along the branches; the differences scatter as expected from the chain statistics, although this check cannot detect an offset that shifts both together.

C. Validation

We benchmark the SSE code against exact diagonalization at finite temperature on four 12-spin periodic clusters built by the production lattice code, so that both

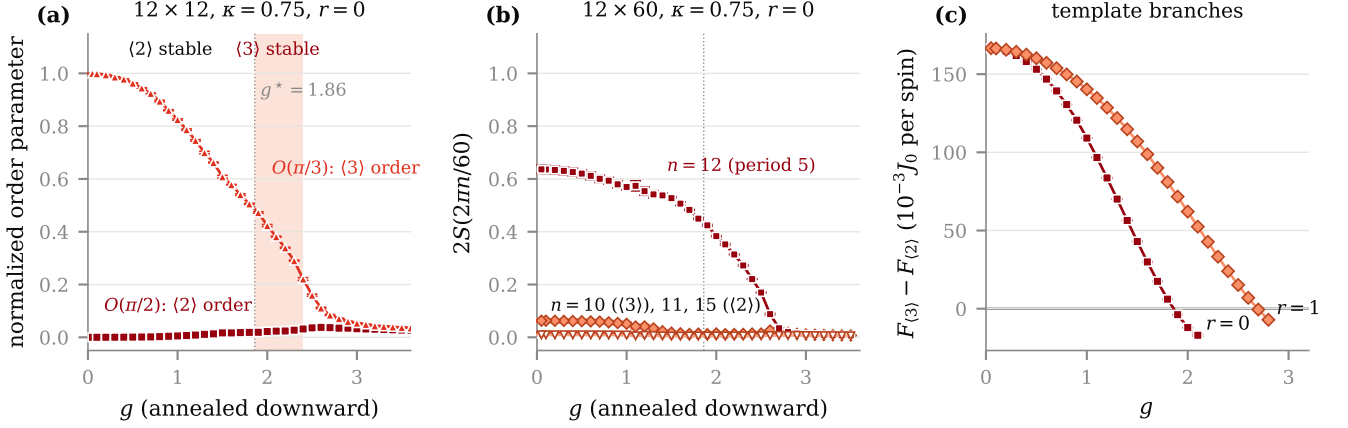


FIG. 3. Annealing freezes at the wavevector at which order forms. (a) Order parameters along anneals from the paramagnet at $\kappa = 0.75$, $r = 0$ on 12×12 bilayers (means of six chains), normalized to their saturation values: $O(\pi/3)$ of $\langle 3 \rangle$ (triangles) and $O(\pi/2)$ of $\langle 2 \rangle$ (squares). The shaded range is where $\langle 3 \rangle$ lies below the FM and $\langle 2 \rangle$ branches, between their crossing g^* (Table IV) and the onset of order. Below g^* the antiphase is lower, yet the chains stay in $\langle 3 \rangle$. (b) The same anneal on 12×60 bilayers, where $\langle 3 \rangle$ ($n = 10$ in $g = 2\pi n/60$), $n = 11$, period 5 ($n = 12$) and $\langle 2 \rangle$ ($n = 15$) are resolved: order forms at $n = 12$, the allowed wavevector nearest the mean-field instability, and stays there. The dotted line marks g^* . (c) $F_{(3)} - F_{(2)}$ from the template branches at $\kappa = 0.75$, which changes sign at g^* .

methods use the same bond list: a 2×3 bilayer, 2×6 and 3×4 single layers, and a 1×6 two-leg ladder, at parameters that include frustrated points up to $\kappa = 1.3$, both signs of $r_\perp$, $J_y < 0$, and $T = 0.15$ and $0.5 J_0$, with longer runs than in production (Table S1 of the Supplemental Material [44]). The 471 comparisons of the energy, m_x , both correlation estimators, template moments and the structure factor scatter slightly less than the Student- t distribution expected for our 24-chain averages [Fig. 6(a)], since comparisons within one parameter set share their chains; the largest $|z|$ is 2.62.

We also test the whole free-energy procedure against exact diagonalization. On two 12-spin clusters, a 2×3 bilayer at $\kappa = 0$, $r = 1$ and a 2×6 layer at $\kappa = 0.3$, chains started in the ferromagnet at g_0 are stepped up to $g = 3$ exactly as the template branches, and m_x is integrated with the same quadrature from the exact free energy at g_0 . On the bilayer the integral reproduces the exact $F(g)$ at every field within 3×10^{-4} . On the frustrated layer, four runs that separate seeds from sweeps agree with the exact values within 4×10^{-4} below the crossover of this small cluster ($g \leq 1.2$); across the crossover two of them, with the same seeds, lie above it by up to 2×10^{-3} . The integration is thus confirmed at the level of 4×10^{-4} in the ordered regime, where all comparisons of this paper lie, while across a crossover occasional errors of order 10^{-3} occur that more sweeps do not reliably remove.

On large lattices the energies of all template branches at g_0 agree with the perturbation series (11) [Fig. 6(b)]. For the two-dimensional transverse-field Ising model ($\kappa = r_\perp = 0$) with $\beta = L$ (dynamic exponent $z = 1$), the crossings of the Binder cumulant U_4 and of ξ_y/L bracket the quantum critical point $g_c = 3.04438(2)$ [47] from below and above, within 0.2% for the largest pair [Fig. 6(c)].

At $T = 0.15 J_0$ the transition lies between $g = 3.0$, where $T_c = 0.2977(9) J_0$ [48], and g_c . Our crossings at this temperature drift downward into this interval but have not converged by $L = 48$, where U_4 at the crossing still rises [Fig. 6(d)]: the crossover from quantum-critical to two-dimensional classical scaling, at $L \approx v\beta$, sets in only at these sizes. For this reason we do not locate the paramagnetic boundary of the full model at $T = 0.15$ in this work.

V. RESULTS

A. The $\langle 3 \rangle$ wedge from branch free energies

We computed template branches for the FM, $\langle 2 \rangle$ and $\langle 3 \rangle$ structures at nine values of κ between 0.47 and 0.75, for $r = 0$ and 1, on 12×12 bilayers at $T = 0.15 J_0$, with six chains per branch and field steps of 0.1, and located their crossings as described in Sec. IV B. Table IV and Fig. 2 compare the crossing fields g^* with the perturbation series evaluated at the actual κ . We use $\lambda = g^*/(2 + r)$ (Sec. III B) to measure how far the series is pushed.

Where the series converges the agreement is quantitative. With the eighth order, the twelve crossings with $\lambda \leq 0.5$ agree with the series with $\chi^2 = 7.2$; the leading order alone is off by a few percent. At the two points with $\lambda \approx 0.49$ for $r = 0$ the truncation error is not controlled, since the last step of the series is larger than the previous one; these crossings lie at the critical field of the rows next to the walls. Without them, $\chi^2 = 6.0$ for the ten remaining crossings, and the slope means of Table V

TABLE IV. Crossing fields of the $\langle 3 \rangle$ branch with the FM branch ($\kappa < 1/2$) or the $\langle 2 \rangle$ branch ($\kappa > 1/2$) on 12×12 bilayers at $T = 0.15 J_0$, with jackknife standard errors, against the crossings of the perturbation series truncated at $O(g^n)$ at the same κ . $\lambda = g^*/(2+r)$; $z_n = (g^* - g_n^*)/\sigma$. For $\kappa > 1/2$ these are crossings of two branches that lie inside the $\langle 23 \rangle$ window, not phase boundaries. Daggers mark the inverse-variance combination of the production run with a finer-grid rerun (production and rerun values: 0.5952(48), 0.6062(13); 0.4378(37), 0.4290(25); 1.1258(49), 1.1200(36); 1.3253(15), 1.3215(22); 1.8597(33), 1.8603(25)). At $\kappa = 0.75$, $r = 0$ the $O(g^8)$ series has no crossing.

κ	r	g^* (QMC)	λ	g_2^*	g_4^*	g_6^*	g_8^*	z_6	z_8
0.47	0	0.9856(32)	0.49	1.0397	0.9766	0.9770	0.9829	+2.6	+0.8
0.48	0	0.8250(38)	0.41	0.8631	0.8247	0.8251	0.8271	0.0	-0.5
0.49	0	0.6054(13) [†]	0.30	0.6210	0.6056	0.6058	0.6060	-0.2	-0.4
0.51	0	0.4318(21) [†]	0.22	0.4408	0.4348	0.4349	0.4349	-1.5	-1.5
0.52	0	0.6008(32)	0.30	0.6149	0.5995	0.5997	0.6000	+0.3	+0.3
0.53	0	0.7190(58)	0.36	0.7432	0.7175	0.7179	0.7187	+0.2	+0.1
0.5625	0	0.9832(48)	0.49	1.0312	0.9713	0.9733	0.9795	+2.1	+0.8
0.625	0	1.3241(13) [†]	0.66	1.3674	1.2525	1.2606	1.2990	+50	+20
0.75	0	1.8601(20) [†]	0.93	1.7550	1.5669	1.5986	—	+132	—
0.47	1	1.5484(26)	0.52	1.5803	1.5327	1.5469	1.5468	+0.6	+0.6
0.48	1	1.3017(41)	0.43	1.3196	1.2918	1.2984	1.2985	+0.8	+0.8
0.49	1	0.9542(57)	0.32	0.9555	0.9449	0.9465	0.9465	+1.4	+1.4
0.51	1	0.6794(51)	0.23	0.6800	0.6757	0.6761	0.6761	+0.7	+0.7
0.52	1	0.9367(57)	0.31	0.9446	0.9336	0.9351	0.9351	+0.3	+0.3
0.53	1	1.1221(29) [†]	0.37	1.1373	1.1187	1.1222	1.1222	-0.0	-0.1
0.5625	1	1.5390(42)	0.51	1.5605	1.5159	1.5307	1.5308	+2.0	+2.0
0.625	1	2.0258(23)	0.68	2.0346	1.9473	1.9941	1.9946	+14	+13
0.75	1	2.6923(48)	0.90	2.5495	2.4079	2.5381	2.5416	+32	+31

change by less than 0.001. Beyond $\lambda = 0.5$ the crossings move above the series [Fig. 2(c)], and the eighth order does not close the gap. We chose the range $\lambda \leq 0.5$ after inspecting Table IV; the edge rows of Sec. III C give it a physical basis, and with their range for $r = 1$, $\lambda \leq 0.61$, the fourteen crossings included still agree ($\chi^2 = 11.6$).

With the wedge boundaries written as $|\kappa - 1/2| = Ag^2 + Bg^4$ and B fixed at its $O(g^4)$ value, each crossing gives an estimate of the leading coefficient A (Table V). The weighted means agree with the identically processed series to about 1% on both sides and for both r , so the $\langle 3 \rangle$ wedge of the lowest-order theory is confirmed, including the factor 2.4 by which the second layer narrows it at fixed g (2.43(2) and 2.41(2) on the two sides; this ratio follows from the slopes and is not an independent test). Excluding the reruns described below changes the means by less than 0.01; extending the window beyond $\lambda = 0.5$ makes the fits on the $\langle 2 \rangle$ side fail.

Two checks bear on the errors. Five cases were rerun with new seeds on finer grids; four were chosen in advance, and $\kappa = 0.49$, $r = 0$ was added because its production value lay 2.2 standard errors below the $O(g^6)$ crossing, so we combine each rerun with its production value rather than replace it. The two differ with $\chi^2 = 9.4$ for five cases after allowing for the Student- t distribution of six-chain errors ($p = 0.09$), a weak hint that these errors are somewhat small; other ways of combining them move the slopes by less than 0.01. Second, 24×24 bilay-

TABLE V. Leading coefficients of the wedge boundaries, $A_i = |\kappa - 1/2|/g^{*2} - Bg^{*2}$ with B fixed at its $O(g^4)$ value, relative to $A_{\text{th}} = F(3)/8$ (FM side) or $F(3)/4$ ($\langle 2 \rangle$ side), for the crossings with $\lambda \leq 0.5$. Last two columns: the same extraction applied to the $O(g^6)$ and $O(g^8)$ crossings of Table IV, which shows the bias of the extraction itself. Last rows: weighted means (with the QMC weights).

Side	r	κ	λ	A_i/A_{th}	$O(g^6)$	$O(g^8)$
FM $\langle 3 \rangle$	0	0.47	0.49	1.008(10)	1.034	1.016
FM $\langle 3 \rangle$	0	0.48	0.41	1.016(12)	1.016	1.009
FM $\langle 3 \rangle$	0	0.49	0.30	1.005(5)	1.004	1.003
$\langle 3 \rangle$ $\langle 2 \rangle$	0	0.51	0.22	1.018(11)	1.002	1.001
$\langle 3 \rangle$ $\langle 2 \rangle$	0	0.52	0.30	1.001(13)	1.006	1.005
$\langle 3 \rangle$ $\langle 2 \rangle$	0	0.53	0.36	1.007(21)	1.012	1.009
$\langle 3 \rangle$ $\langle 2 \rangle$	0	0.5625	0.49	1.007(15)	1.039	1.019
FM $\langle 3 \rangle$	1	0.48	0.43	1.001(8)	1.007	1.007
FM $\langle 3 \rangle$	1	0.49	0.32	0.983(13)	1.002	1.002
$\langle 3 \rangle$ $\langle 2 \rangle$	1	0.51	0.23	0.990(16)	1.001	1.001
$\langle 3 \rangle$ $\langle 2 \rangle$	1	0.52	0.31	1.000(15)	1.004	1.004
$\langle 3 \rangle$ $\langle 2 \rangle$	1	0.53	0.37	1.008(7)	1.008	1.008
FM $\langle 3 \rangle$	0	mean		1.007(4)		
$\langle 3 \rangle$ $\langle 2 \rangle$	0	mean		1.010(7)		
FM $\langle 3 \rangle$	1	mean		0.996(7)		
$\langle 3 \rangle$ $\langle 2 \rangle$	1	mean		1.005(6)		

ers at five of the points ($\kappa = 0.47$ and 0.53 for both r , and $\kappa = 0.625$ for $r = 0$) reproduce the 12×12 crossings within 0.24%.

B. The $\langle 23 \rangle$ phase

We tested the fourth-order prediction with simulations whose design, analysis and pass criteria were fixed before the runs (Appendix C). At seven points with $\lambda = g/(2+r) \leq 0.5$, four for $r = 0$ and three for $r = 1$, we followed $\langle 23 \rangle$ and $\langle 2 \rangle$ on 20×20 bilayers and $\langle 23 \rangle$ and $\langle 3 \rangle$ on 20×30 bilayers, so that each comparison is made on one lattice, with 12 chains per structure. The test field g_c is the $O(g^6)$ crossing of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches, where the series predicts the largest margin for $\langle 23 \rangle$. A point passes if $\langle 23 \rangle$ lies below both neighbors at g_c by more than two standard errors (one-sided Welch t) and no chain has left its structure.

$\langle 23 \rangle$ passes at six of the seven points, mostly by a wide margin (Table VI, Fig. 4). Its gain over $\langle 2 \rangle$ agrees with the series at every point, at sixth as at eighth order (Table VII). Its gain over $\langle 3 \rangle$ exceeds the sixth-order prediction at most points; for decoupled layers the eighth order accounts for this, in the bilayer it does not (see below). The one failure is the point with the smallest predicted margin, $\kappa = 0.52$ at $r = 1$, where $\langle 23 \rangle$ lies below $\langle 2 \rangle$ by 1.8 standard errors against the required 2.12, although the measured gain agrees with the prediction. With the realized errors, passing all fourteen comparisons had a probability of only 0.58 even if the theory is exact (Appendix C). A follow-up at this point with 24 new chains per structure, decided after the result and reported separately, resolves $\langle 23 \rangle$ ($t = -4.6$ and -7.5 ; Table S3 of the Supplemental Material [44]); the pooled values of Fig. 5, chosen after the fact, carry no valid p -value.

The lower edge of the window agrees with the series at every point (Fig. 5). The upper edge lies above the sixth-order value at six of the seven points, by more than 2σ at four, including both points at $\kappa = 0.53$, where the pre-registration anticipated agreement within 2σ . For decoupled layers the eighth order removes most of this excess; in the bilayer the runs described next trace it to the pre-registered 20×30 runs. At $\kappa = 0.5625$ the upper edge is not the end of the phase: $\langle 233 \rangle$ drops below $\langle 23 \rangle$ first (Sec. VC). The ratio of the window widths for $r = 0$ and $r = 1$ agrees with the eighth-order series [Fig. 5(c)]; the second layer narrows the $\langle 23 \rangle$ window several times more strongly than the $\langle 3 \rangle$ wedge at fixed g .

Further runs, decided after the test, check these results; the Supplemental Material [44] (Sec. S3 and Table S3) gives them in full, with the error budget. On a common 20×60 lattice, for both r and with two independent runs in the bilayer, both edges and both gains agree with the eighth-order series within 2σ , and at $\kappa = 0.52$, $r = 1$ the narrowest pre-registered margins become clear ones: $\langle 23 \rangle$ lies $1.6\text{--}1.9 \times 10^{-4}$ per spin below $\langle 3 \rangle$ (five to nine standard errors) and 1.6×10^{-4} below $\langle 2 \rangle$ (6.5 stan-

dard errors). The one discrepancy is in the pre-registered bilayer comparisons with $\langle 3 \rangle$ on 20×30 lattices at the standard number of sweeps, which favor $\langle 23 \rangle$ by about 10^{-4} per spin too much; with four times the sweeps they agree with the common lattice and with the series, and we think those runs were not fully equilibrated. Runs repeated with identical settings and new seeds agree within their statistical errors, $\langle 23 \rangle$ remains below both neighbors at $T = 0.10 J_0$, and 40×40 bilayers reproduce the lower edge. The statistical errors, $0.3\text{--}1.2 \times 10^{-4}$ per spin, dominate the error budget except for $r = 0$ at $\kappa \geq 0.545$, where the truncation of the series and thermal excitations of the rows next to the walls reach several 10^{-4} and the comparison of the upper edge with the series is qualitative. $\langle 23 \rangle$ lies below both neighbors in every run and on every lattice at all seven points, the energies at g_0 agree with the series [Fig. 6(b)], and no chain left its structure before g_c .

Structures that the theory places just above $\langle 23 \rangle$ at g_c , namely $\langle 223 \rangle$, $\langle 233 \rangle$ and $\langle 2333 \rangle$ (on 20×28 , 20×32 and 20×22), lie above it at the four points where we followed them, by amounts that agree with the series within about 2σ once the eighth order is included.

C. Beyond the perturbative regime

All fields analysed here lie inside the ordered region: along the anneals on 12×12 lattices order sets in at $g \approx 2.2\text{--}2.4$ for $r = 0$ and $3.1\text{--}3.4$ for $r = 1$, above every crossing of Table IV. We do not locate the paramagnetic boundary itself (Sec. IV C).

Beyond $\lambda = 0.5$ the crossings move increasingly above the series (Table IV), and the eighth order does not close the gap. The bilayer at $\kappa = 0.75$ shows what this means for the phase diagram. The leading-order wedge expanded about $\kappa = 1/2$, Eq. (9), places the $\langle 3 \rangle|\langle 2 \rangle$ crossing at $g = 3.46$, where order sets in, and would leave no field range for $\langle 3 \rangle$; in the simulations $\langle 3 \rangle$ is the lowest of the structures compared from $g^* = 2.69$ up to the onset of order.

Above the $\langle 23 \rangle$ window the simulations point to further structures [Fig. 4(c)]. At $\kappa = 0.5625$ and $r = 0$, $\langle 233 \rangle$ drops below $\langle 23 \rangle$ before $\langle 3 \rangle$ does (Table VI), and $\langle 2333 \rangle$ is lower still from $g \approx 1.3$, so with increasing field the lowest of the structures we followed is $\langle 2 \rangle$, $\langle 23 \rangle$, $\langle 233 \rangle$, $\langle 2333 \rangle$. These fields lie at $\lambda = 0.53\text{--}0.71$, beyond the range of the series. The sixth-order series places the $\langle 23 \rangle|\langle 233 \rangle$ crossing close to the measured one, but the eighth order moves it above $g = 1.6$, so this agreement is fortuitous; at $g = 1.2$ the series fails by about 10^{-3} per spin. Closer to the multiphase point the effect fades: at $\kappa = 0.545$ and 0.53 the range in which $\langle 233 \rangle$ is lowest is narrower than its uncertainty, and in the bilayer at $\kappa = 0.53$ $\langle 3 \rangle$ overtakes $\langle 23 \rangle$ before $\langle 233 \rangle$ does.

We report these structures as an indication only. Only five structures were compared, so a longer structure could be lower still. The gaps at $g = 1.2$ are several times the

TABLE VI. Pre-registered test of the $\langle 23 \rangle$ phase, at $\lambda = g_c/(2+r)$ between 0.30 and 0.49. ΔF_2 and ΔF_3 are the free-energy differences per spin between $\langle 23 \rangle$ and $\langle 2 \rangle$ (on 20×20) and between $\langle 23 \rangle$ and $\langle 3 \rangle$ (on 20×30) at g_c , in units of $10^{-4} J_0$, with standard errors and Welch t values. ΔF_{th} is the $O(g^6)$ prediction, the same for both at g_c . $g_{2|23}$ and $g_{23|3}$ are the edges of the window where $\langle 23 \rangle$ is below $\langle 2 \rangle$ and below $\langle 3 \rangle$, next to their $O(g^6)$ values. $w = \Delta\kappa/g_c^4$ (units of 10^{-3}) is the width of the window converted to κ with the slope of the measured $\langle 3 \rangle|\langle 2 \rangle$ line.

κ	r	g_c	ΔF_2 (t)	ΔF_3 (t)	ΔF_{th}	$g_{2 23}$ (th)	$g_{23 3}$ (th)	w (th)	Test
0.52	0	0.5997	-3.32(60) (-5.6)	-1.46(62) (-2.4)	-2.17	0.5885(21) (0.5923)	0.6084(36) (0.6123)	11.3(2.4) (11.4)	pass
0.53	0	0.7179	-4.93(76) (-6.5)	-7.19(99) (-7.3)	-4.98	0.7050(19) (0.7049)	0.7553(51) (0.7418)	17.8(1.9) (13.4)	pass
0.545	0	0.8525	-12.13(82) (-14.9)	-14.96(100) (-14.9)	-11.42	0.8291(16) (0.8302)	0.9256(52) (0.8990)	22.4(1.3) (16.6)	pass
0.5625	0	0.9733	-23.04(86) (-26.8)	-30.65(122) (-25.2)	-22.19	0.9384(13) (0.9396)	1.1605(91) (1.0561) ^a	37.6(1.6) (21.2)	pass
0.52	1	0.9351	-0.99(54) (-1.8)	-2.47(58) (-4.2)	-1.75	0.9297(30) (0.9254)	0.9577(52) (0.9512)	1.72(37) (1.58)	fail ^b
0.53	1	1.1222	-4.31(66) (-6.5)	-4.90(34) (-14.5)	-3.97	1.1037(29) (1.1051)	1.1604(27) (1.1525)	2.17(15) (1.81)	pass
0.545	1	1.3369	-8.65(91) (-9.5)	-9.57(71) (-13.4)	-8.98	1.3082(30) (1.3070)	1.4020(48) (1.3951)	2.34(14) (2.20)	pass

^aAbove $g = 1.064(15)$ the structure $\langle 233 \rangle$ lies below $\langle 23 \rangle$ (Sec. V C). ^bPasses in a follow-up with 24 new chains per structure, decided after this result (Sec. V B).

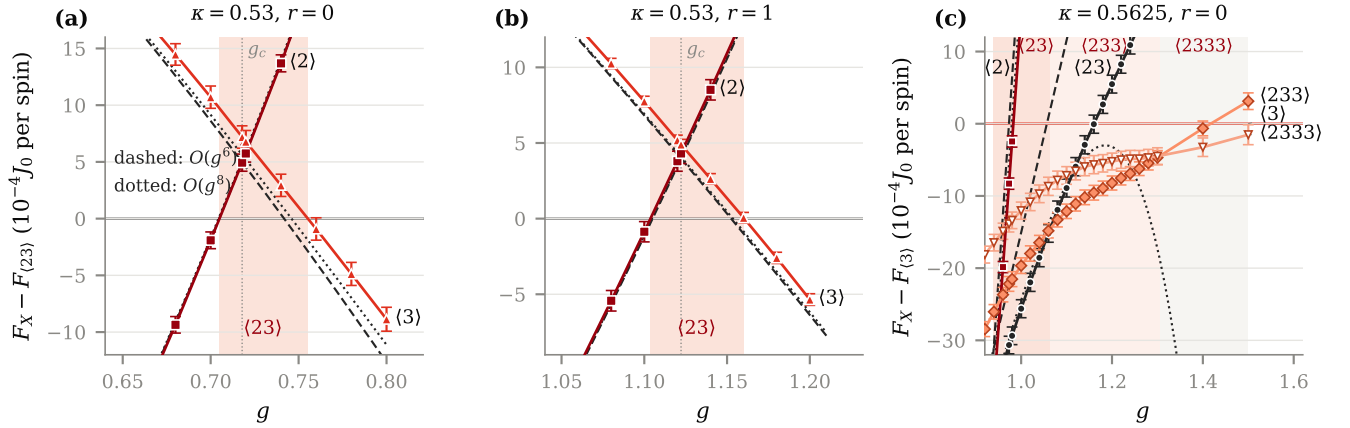


FIG. 4. Free energies of competing structures near the $\langle 23 \rangle$ window. (a), (b) $F_X - F_{\langle 23 \rangle}$ at $\kappa = 0.53$ for $r = 0$ and 1, for $X = \langle 2 \rangle$ (squares, 20×20 bilayers) and $X = \langle 3 \rangle$ (triangles, 20×30), each against $\langle 23 \rangle$ on the same lattice. $\langle 23 \rangle$ is stable where both are positive (shaded, edges from the simulations). Dashed lines: $O(g^6)$, dotted lines: $O(g^8)$, both at the actual κ . g_c is the field of the pre-registered test. (c) $\kappa = 0.5625$, $r = 0$: free energies relative to $\langle 3 \rangle$ (20×30), including $\langle 233 \rangle$ (20×32 , diamonds) and $\langle 2333 \rangle$ (20×22 , open triangles); $\langle 23 \rangle$ is shown as circles. The lowest curve is the stable structure among those followed, and the shaded ranges mark where $\langle 23 \rangle$, $\langle 233 \rangle$ and $\langle 2333 \rangle$ are lowest. In (b) the differences to $\langle 3 \rangle$ (20×30 , standard sweeps) carry a bias of about -10^{-4} (Sec. V B). Error bars are one standard error.

TABLE VII. The $O(g^8)$ predictions at the seven test points of Table VI, computed after the test: the free-energy differences ΔF_2 and ΔF_3 at the pre-registered g_c , which is the $O(g^6)$ crossing of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches (in $10^{-4} J_0$), and the window edges. At $\kappa = 0.5625$, $r = 0$ the eighth-order series has no $\langle 23 \rangle|\langle 3 \rangle$ crossing below $g = 1.6$.

κ	r	$\Delta F_2^{(8)}$	$\Delta F_3^{(8)}$	$g_{2 23}^{(8)}$	$g_{23 3}^{(8)}$
0.52	0	-2.21	-2.32	0.5922	0.6134
0.53	0	-5.16	-5.63	0.7045	0.7460
0.545	0	-12.12	-14.01	0.8291	0.9176
0.5625	0	-24.18	-29.74	0.9373	—
0.52	1	-1.78	-1.78	0.9253	0.9515
0.53	1	-4.08	-4.09	1.1046	1.1536
0.545	1	-9.42	-9.44	1.3058	1.3994

bias of about 10^{-4} found in some runs at the standard sweeps (Sec. V B), but the $\langle 233 \rangle|\langle 2333 \rangle$ crossing is not robust to a shift of that size, which would move it by about 0.08. The walls may also soften at these fields (Sec. III C), which the template branches would not capture. And the test was exploratory (T8 in Appendix C). We therefore regard $\langle 233 \rangle$ and $\langle 2333 \rangle$ as the lowest of the structures we followed, not as established phases.

Near the paramagnet the order may become incommensurate or floating, as in the classical two-dimensional model [5, 8]. This regime lies beyond our lattices and templates (Sec. VI).

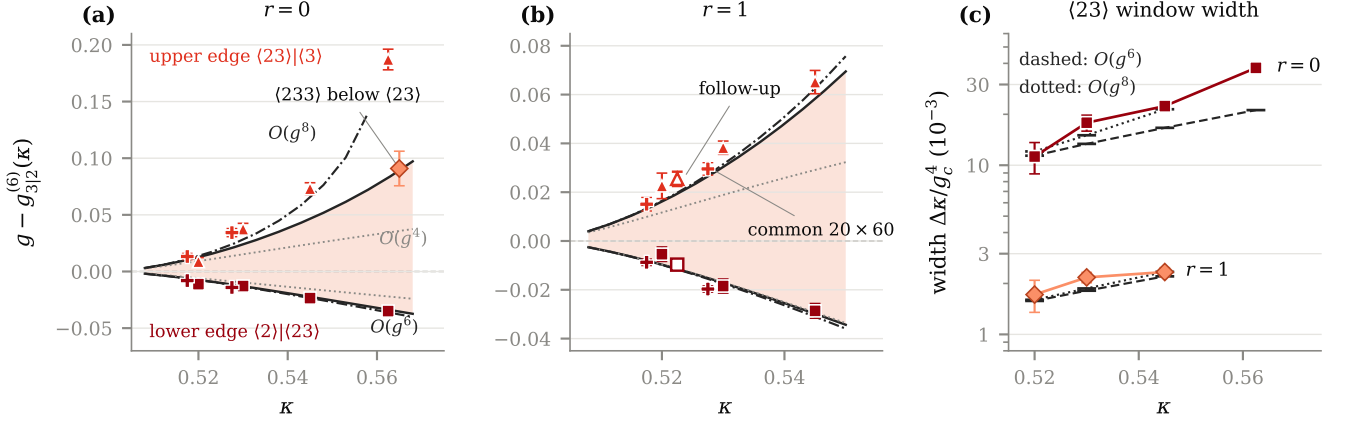


FIG. 5. Edges of the $\langle 23 \rangle$ window at the seven test points for (a) $r = 0$ and (b) $r = 1$, measured from the $O(g^6)$ crossing $g_{3|2}^{(6)}(\kappa)$ of the $\langle 3 \rangle$ and $\langle 2 \rangle$ branches. Squares: lower edge, where $\langle 23 \rangle$ drops below $\langle 2 \rangle$; triangles: upper edge, where $\langle 3 \rangle$ drops below $\langle 23 \rangle$. Solid lines and shading: $O(g^6)$; dotted lines: $O(g^4)$; dash-dotted lines: $O(g^8)$, whose upper edge for $r = 0$ ends where the eighth-order series stops crossing. Open symbols at $\kappa = 0.52$, $r = 1$: the follow-up pooled with the original chains (36 per structure), decided after the pre-registered result. Plus signs: edges with $\langle 2 \rangle$, $\langle 23 \rangle$ and $\langle 3 \rangle$ on one 20×60 lattice; for $r = 1$ the chains of two independent runs are pooled. Both are displaced slightly in κ for clarity. The diamond at $\kappa = 0.5625$, $r = 0$ marks where $\langle 233 \rangle$ drops below $\langle 23 \rangle$ (Sec. V C). (c) Widths $w = \Delta\kappa/g_c^4$ of the windows (Table VI) against $O(g^6)$ (dashed) and $O(g^8)$ (dotted; no value at $\kappa = 0.5625$, $r = 0$), on a logarithmic scale. For $r = 1$ the widths use the pre-registered upper edges, which the bias of Sec. V B raises by about 0.008 in g , 10–30% of the width.

VI. DISCUSSION AND CONCLUSION

For spin 1/2 in a transverse field on the square lattice, quantum fluctuations select $\langle 3 \rangle$ at the multiphase point at order g^2 . The mechanism is local: spins next to a straight domain wall have the smallest molecular field and gain the most energy from virtual flips, and domains of three sites carry the largest number of such spins per site among domains that avoid the two-site penalty. The interlayer coupling raises every local field by the same amount and weakens the selection. At the $\langle 3 \rangle|\langle 2 \rangle$ boundary, fourth, sixth and eighth order then select $\langle 23 \rangle$, $\langle 223 \rangle$ and $\langle 2223 \rangle$. The simulations confirm the two structures that are wide enough to resolve, for $\lambda \lesssim 0.5$: the slope of the $\langle 3 \rangle$ wedge to about 1%, and, in post-hoc runs on a common lattice at four points, both edges of the $\langle 23 \rangle$ window within their statistical errors. $\langle 23 \rangle$ is the lowest structure at all seven test points in every run.

The order of appearance is that of the Fisher–Selke analysis of the classical three-dimensional model [2, 3, 27], with g^2 in place of the low-temperature expansion parameter (Sec. III C); Ref. [49] compares thermal and quantum splitting of the multiphase point in the large-spin model. Unlike the large-spin Heisenberg analog, where spin deviations propagate between walls [17, 18], and the transverse chain, where mobile walls produce a floating phase already at first order in g [12], the walls here are rigid: a transverse field creates only local flips, so the order g^{2n} couples at most n consecutive domains. The same rigidity limits the series, because the rows next to the walls are themselves transverse-field chains or ladders, which become critical at $g = 1$ ($r = 0$) and $g \approx 1.83$

($r = 1$). The interlayer coupling suppresses the phases that appear at higher order more strongly than $\langle 3 \rangle$. Between $r = 0$ and $r = 1$ the $\langle 3 \rangle$ wedge narrows by 2.4 and the $\langle 23 \rangle$ window by 6.7 at fixed g ; most of this reflects the larger local fields, and at fixed λ the factors are 1.07, 1.32, 1.58 and 1.86 for $\langle 3 \rangle$, $\langle 23 \rangle$, $\langle 223 \rangle$ and $\langle 2223 \rangle$.

We establish the first four members of the sequence, and the work points to several next steps. (i) Higher orders of the series would show whether further phases $\langle 2^k 3 \rangle$ follow on the $\langle 2 \rangle$ side, as in the classical problem, and would decide the $\langle 23 \rangle|\langle 223 \rangle$ edge, which is still degenerate at order g^8 . (ii) Above the $\langle 23 \rangle$ window, beyond the range of the series, the simulations find $\langle 233 \rangle$ and then $\langle 2333 \rangle$ below $\langle 23 \rangle$ and $\langle 3 \rangle$ at $\kappa = 0.5625$. The template method compares only the structures we follow, so a systematic search over longer sequences would identify the stable structure there. (iii) On 20×30 bilayers at the standard number of sweeps the gain of $\langle 23 \rangle$ over $\langle 3 \rangle$ comes out about $10^{-4} J_0$ per spin larger than with four times the sweeps or on 20×60 lattices; understanding why these runs equilibrate more slowly would sharpen the error model of the template method. (iv) Near the paramagnetic boundary the order may become floating, as in the classical two-dimensional model [5, 8]. Locating this boundary calls for larger lattices and a finite-size analysis, which our test on the transverse-field Ising model shows to be demanding at this temperature.

Annealing ten times more slowly gives the same result on 12×12 lattices, so the walls do not relax at the rates we can afford; we used one update scheme throughout, and other schemes may fare differently. Annealing from the paramagnet freezes at the wavevector at which order

first develops, the allowed wavevector nearest the instability of the paramagnet, and the rigid walls then prevent any reorganization at lower field. On 12×12 lattices this selects $\langle 3 \rangle$ at every κ we studied, including where the ferromagnet or $\langle 2 \rangle$ is stable. On 12×60 lattices at $\kappa = 0.625$ and 0.75 it selects sequences of two- and three-site domains with the corresponding wall densities. In the bilayer at $\kappa = 0.75$, where the leading-order wedge, expanded about $\kappa = 1/2$, would have placed the $\langle 3 \rangle|\langle 2 \rangle$ crossing at the ordering field, no anneal reaches $\langle 2 \rangle$, although $\langle 2 \rangle$ is lower than $\langle 3 \rangle$ below $g^* = 2.69$. Similar obstacles may affect annealing protocols on programmable quantum devices that realize frustrated transverse-field Ising models [26].

ACKNOWLEDGMENTS

We thank Can Tho University for its support. Claude Opus 5.5 (Anthropic) was used throughout the preparation of this work: to search and survey the literature, to write and run the simulation and analysis code, to check the perturbative coefficients independently, and to check the numerical claims and the characterizations of prior work against the data and sources.

DATA AVAILABILITY

The data that support the findings of this article, including the branch free energies, the exact perturbative coefficients and the analysis scripts, are available from the authors upon reasonable request.

Appendix A: Minimum of the domain-sequence energy

Write the per-domain cost as $c(n) = (2 - 4\kappa) - \frac{g^2}{2}F(n)$, so that $\Delta e = \sum_j c(n_j)/\sum_j n_j$ by Eq. (7). For positive n_j , $\sum_j c(n_j)/\sum_j n_j \geq \min_j c(n_j)/n_j$, with equality when every domain has the same ratio $c(n_j)/n_j$. If $c(n) > 0$ for all n , the minimum is the ferromagnet, $\Delta e = 0$. Otherwise it is reached by the periodic structure that minimizes $c(n)/n$. From Eq. (8),

$$F(3) - F(n \geq 4) = \frac{2}{(3+r)(4+r)(5+r)} > 0. \quad (\text{A1})$$

With F constant for $n \geq 4$, $\langle n \rangle$ with $n \geq 4$ can beat both the ferromagnet and $\langle 3 \rangle$ only if $F(n) > F(3)$, which never holds. The remaining comparisons, FM against $\langle 3 \rangle$ ($c(3) < 0$) and $\langle 3 \rangle$ against $\langle 2 \rangle$ ($c(3)/3 < c(2)/2$), give Eq. (9). At $\kappa_{3|2}$ one has $c(2)/2 = c(3)/3$, and any sequence of two- and three-site domains has the same Δe .

Higher orders. At order g^4 on the line $k = F(3)/4$ the same ratio structure appears, with the domain costs replaced by costs of adjacent domain pairs. Let

$P(D, D', B) = -4B(D + D')/[(DD')^2(D + D' - 4B)]$ be the pair term of Eq. (12). Let $D_1, \dots, D_n$ be the flip costs along a domain (Table I), and let $\gamma_i = s_i(s_{i+2} + s_{i-2})$, which takes the values $-2, 0$ and 2 . Let X_n and Y_n be the flip costs of the first and second site from a wall of a domain of length n . Then, for a period of L sites,

$$\begin{aligned} W_4 L &= \sum_j [a(n_j) + b(n_j, n_{j+1})], \\ a(n) &= \sum_{i=1}^n \left[-\frac{2k\gamma_i}{D_i^2} + \frac{1}{D_i^3} \right] \\ &\quad + \sum_{i=1}^n \left[P(D_i, D_i, 1) + \frac{1}{2}P(D_i, D_i, r) \right] \\ &\quad + \sum_{i=1}^{n-1} P(D_i, D_{i+1}, 1) + \sum_{i=1}^{n-2} P(D_i, D_{i+2}, -\frac{1}{2}), \\ b(n, n') &= P(X_n, X_{n'}, -1) + P(Y_n, X_{n'}, \frac{1}{2}) \\ &\quad + P(X_n, Y_{n'}, \frac{1}{2}). \end{aligned} \quad (\text{A2})$$

With $C(n, n') = k_1(n - 4) + a(n) + b(n, n')$, the fourth-order energy per spin is $\sum_j C(n_j, n_{j+1})/\sum_j n_j$. A periodic sequence is a closed walk on the directed graph with vertices $\{2, 3\}$ and all four edges, where edge $n \rightarrow n'$ has cost $C(n, n')$ and length n . Every closed walk splits into elementary cycles. A ratio of sums with positive denominators is at least the smallest ratio of its parts, with equality only if all parts share that ratio. The elementary cycles are $2 \rightarrow 2$, $3 \rightarrow 3$ and $2 \rightarrow 3 \rightarrow 2$, that is $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$, so one of them minimizes the energy for every k_1 .

Now count the adjacent pairs N_{22} , N_{33} and $N_{23} = N_{32}$ of a cyclic sequence. The period is $L = 2N_{22} + 3N_{33} + 5N_{23}$, and the total cost is $N_{22}C(2, 2) + N_{33}C(3, 3) + N_{23}[C(2, 3) + C(3, 2)]$. The point (ρ, W_4) of the sequence is therefore the average of the points of $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$, weighted by the site fractions $2N_{22}/L$, $3N_{33}/L$ and $5N_{23}/L$. Sequences with $N_{33} = 0$ lie on the segment from $\langle 2 \rangle$ to $\langle 23 \rangle$, and sequences with $N_{22} = 0$ lie on the segment from $\langle 23 \rangle$ to $\langle 3 \rangle$. All other sequences lie strictly above the hull, because $\langle 23 \rangle$ lies strictly below the chord from $\langle 3 \rangle$ to $\langle 2 \rangle$.

The argument rests on a locality property. Domains have at least two sites and every coupling spans at most two sites, so a connected cluster of n spins touches at most n consecutive domains, and at $\kappa = 1/2$ the field of a spin depends only on its own domain (Table I). The energy at order g^{2n} is therefore a sum of terms for n consecutive domains. Non-periodic sequences are covered as limits of long periodic segments, whose energy obeys the same bound. At order g^6 the terms of Eq. (14) involve at most three consecutive domains. The same argument, applied to the graph whose vertices are pairs of consecutive domains, leaves $\langle 2 \rangle$, $\langle 23 \rangle$ and $\langle 223 \rangle$ as the only candidates on the $\langle 23 \rangle|\langle 2 \rangle$ line, and $\langle 3 \rangle$, $\langle 23 \rangle$ and $\langle 233 \rangle$ on the $\langle 3 \rangle|\langle 23 \rangle$ line.

At order g^8 the terms involve at most four consecutive domains, and the argument applies to the graph

whose vertices are triples of consecutive domains. On the $\langle 223 \rangle | \langle 2 \rangle$ line the sequences degenerate at order g^6 have isolated three-site domains separated by at least two two-site domains, and their elementary cycles are $\langle 2 \rangle$, $\langle 223 \rangle$ and $\langle 2223 \rangle$; longer sequences such as $\langle 22223 \rangle$ or $\langle 2232223 \rangle$ decompose into these and lie on the segments between them. The explicit evaluation places $\langle 2223 \rangle$ below the chord, so it is the one structure that enters this line at order g^8 (Sec. III C). On the $\langle 23 \rangle | \langle 223 \rangle$ line the sequences degenerate at order g^6 are built from the units 23 and 223. Their triples of consecutive domains are (2, 3, 2), (3, 2, 3), (3, 2, 2) and (2, 2, 3), and the only elementary cycles are $\langle 23 \rangle$ and $\langle 223 \rangle$, both through (2, 3, 2). Every such sequence therefore lies on the segment between them at order g^8 , and no structure enters this line at this order.

Appendix B: SSE implementation and validation

Bonds are grouped into classes with a common $|J_b|$ (bonds along x , bonds along y , next-nearest-neighbor bonds along y , rungs). An insertion first chooses a class with probability proportional to its total weight and then a bond uniformly within it, and is rejected when the chosen bond is unsatisfied. The string length grows as $M = n + n/2 + 64$ during thermalization. Each chain starts with 200 diagonal-only sweeps at its first field. Without them, the first cluster update on a nearly empty string would flip every operator-free spin with probability 1/2 and randomize the initial structure. Diagonal updates cannot change the spins, so this step is part of the thermalization. Every later update satisfies detailed balance with respect to the SSE weight, so a chain samples that weight restricted to the basin of its initial structure for as long as it stays in it; chains are cut when their dominant wavevector changes. The diagonal-only sweeps only place a chain in that basin.

The protocol and the full comparison of the exact-diagonalization benchmark are given in the Supplemental Material [44] (Table S1).

Figure 6 summarizes the checks of Sec. IV C.

Appendix C: Pre-registered test of the $\langle 23 \rangle$ phase

The test of Sec. VB followed a written plan. Its design, analysis and pass criteria were fixed before the production runs, after an independent review of the design; the full plan, the outcomes of all its criteria and the change log are given in the Supplemental Material [44] (Sec. S2 and Table S2). At each of seven points (κ, r) with $\lambda \leq 0.5$, the pairs $\{\langle 23 \rangle, \langle 2 \rangle\}$ ran on a 20×20 bilayer and $\{\langle 23 \rangle, \langle 3 \rangle\}$ on a 20×30 bilayer, with 12 chains per structure and $2000 + 3000$ sweeps per field at $T = 0.15 J_0$. The criteria used in the main text are (T1) $F_{\langle 23 \rangle} < F_{\langle 2 \rangle}$ and $F_{\langle 23 \rangle} < F_{\langle 3 \rangle}$ at g_c , each with one-sided Welch $p < 0.0228$ and no chain cut before g_c , required

at all seven points; (T6) consistency of $\langle 23 \rangle$ on the two lattices, $\max |z| < 3$; and (T8) an exploratory search for further structures above the window. The stop rule, no claim for $\langle 23 \rangle$ if a competitor lay more than 3σ below it at g_c , was never triggered. At $\kappa = 0.53$, $r = 1$, T6 passed its criterion, but with $\chi^2 = 3.45$ per point and 14 of 25 fields beyond 2σ ; the original 20×30 run for $\langle 23 \rangle$ at this point is the one that the repeat with four times the sweeps did not reproduce (Sec. VB).

Power, computed after the test. With the planned errors the probability of passing all fourteen comparisons, if the theory is exact, was between 0.5 and 0.7; with the realized errors it was 0.58. The headline criterion was therefore underpowered, and it was not met.

The follow-up at $\kappa = 0.52$, $r = 1$ and the common-lattice runs were decided after the result of T1 was known, and are reported separately.

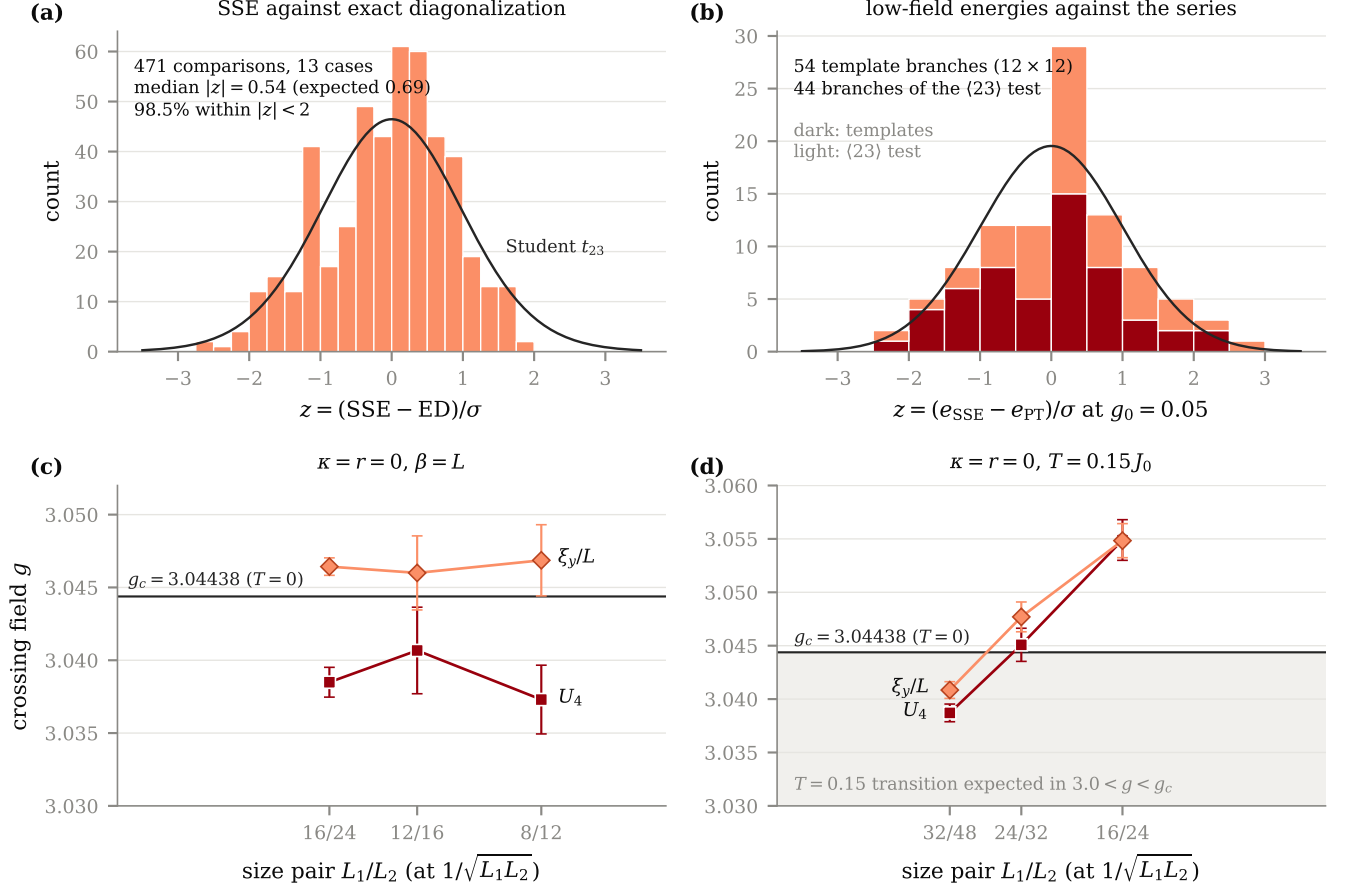


FIG. 6. Validation. (a) Pulls z of all 471 comparisons of SSE with exact diagonalization (Table S1 of the Supplemental Material [44]), against the Student- t distribution with 23 degrees of freedom expected for averages of 24 chains. (b) Pulls of the measured energy at $g_0 = 0.05$ against the perturbation series, for the 54 template branches of Sec. V A (dark) and the 44 branches of the $\langle 23 \rangle$ test (light), with the standard normal density. (c), (d) Two-dimensional transverse-field Ising model: crossing fields of the Binder cumulant U_4 and of ξ_y/L for pairs of sizes, (c) with $\beta = L$ and (d) at $T = 0.15 J_0$. The horizontal line is the quantum critical point $g_c = 3.04438(2)$ [47]. At $T = 0.15 J_0$ the transition lies between $g = 3.0$ [48] and g_c (shaded).

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